Harsh Karan

Software Engineer

Email: harshkaran163@gmail.com Phone: +91 9140804762 LinkedIn: linkedin.com/in/harsh-karan GitHub: github.com/HarshKaran

CORE COMPETENCIES

Full Stack Development • React • TypeScript • Python • FastAPI • Node.js • REST APIs • Data Visualization (Recharts, Chart.js) • Analytics Dashboards • SQL & Relational Databases • State Management • Component Design Systems • Agile / Scrum • Testing • AI Coding Tools (Cursor, Claude Code, Copilot) • CI/CD

EXPERIENCE

Software Engineer 1 | **MAQ Software** Jan 2025 – Present

- Built **full-stack analytics interfaces** using **React, TypeScript, and Node.js**, developing interactive BI dashboards with charting libraries (**Recharts, Chart.js**) for real-time usage monitoring, dataset querying, and decision-support views used by internal business users.
- Developed **backend REST APIs** in **Python (FastAPI)** and Node.js/Express.js with clean request/response contracts, serving usage analytics based on business logic and improving analytics request **throughput by ~40%** through optimized query handling and response caching.
- Wrote **SQL queries and data-access logic** against relational schemas, debugging query issues and optimizing database performance to support data-heavy analytics workloads.
- Built a production-grade **Retrieval-Augmented Generation (RAG)** solution with React, Node.js, and LLM technologies, delivering context-aware responses on an internal portal to support business decision-making.
- Architected an **AI-driven multi-agent orchestration framework** that automates complex business and DevOps workflows across enterprise platforms, delivering **13.6 hours/week** of productivity savings across **14 production workflows**.
- Delivered features across the full **SDLC** — development, testing, and deployment — within an **Agile/Scrum** team, writing tests for backend and frontend code and participating in **structured feedback sessions** with stakeholders; used **Cursor, Claude Code, and Copilot** to accelerate development.
- Engineered an **automated validation framework** using React, Node.js, and Playwright to run sanity checks across **90-page reports**, cutting manual validation effort from **3 hours to 30–45 minutes** and auto-publishing consolidated **HTML reports** to Teams channels.

Software Engineer | **MediaTek** May 2024 – Dec 2024

- Developed responsive components using React.js, Tailwind CSS, and JavaScript for internal tooling and platform enhancements.
- Created dashboards for MediaTek Dimensity 9400 and Imagiq systems, improving visualization and debugging accuracy.
- Engineered a 360° virtual-tour platform for scene uploading, rendering, and cross-chipset optimization.

EDUCATION

Bundelkhand Institute of Engineering and Technology, Jhansi 2020 – 2024
B.Tech in Information Technology — CGPA: 7.67

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++
Frontend: React, Redux, TypeScript, Recharts, Chart.js, Three.js, Tailwind CSS, HTML/CSS
Backend: Python (FastAPI), Node.js, Express.js, REST APIs, MCP Servers
Databases: SQL (MySQL, SQL Server), Schema Design, Query Optimization, MongoDB
AI & Automation: Agentic AI, Multi-Agent Orchestration, RAG, LLM Integration
AI Coding Tools: Cursor, Claude Code, GitHub Copilot
Cloud & DevOps: Azure, Azure DevOps, CI/CD, Git/GitHub
Practices: Agile / Scrum, SDLC, Unit & Integration Testing, Troubleshooting, Code Review

SELECTED PROJECTS

Sandesha | github.com/HarshKaran/Sandesha
Built a full-stack real-time communication platform with chat, posts, audio/video calling, and presence indicators using **React, Node.js, Socket.IO, WebRTC, Redis, and MongoDB**, delivering a responsive, user-focused product experience.

CODING PROFILES

Leetcode GeekForGeeks

CERTIFICATIONS & ACHIEVEMENTS

- **Global Rank 58** — CodeChef December Long Challenge 2022 (Div-3), demonstrating strong problem-solving and debugging skills.
- Solved **600+ coding problems** across LeetCode, GeeksforGeeks, and CodeStudio.
- Microsoft Certified: DP-600, DP-700, DP-800, AI-103.